

# AUSTIN TONSAY

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## **EDUCATION:**

**California State Polytechnic University, Pomona**

*Expected Graduation: May 2029*

Bachelor of Science, Aerospace Engineering (Emphasis: Astronautics) | GPA: 3.83

## **PERSONAL PROJECTS:**

**Baseline vs. Altitude-Optimized High-Power Rocketry Experiment:**

*May 2026 – Present*

- Performing aerodynamic/stability trades across fin, nose, body, and boattail geometry and surface finish in SolidWorks/OpenRocket to design baseline and altitude-optimized high-power rockets predicted to reach 3,095 ft and 4,523 ft (~46% increase)
- Iterating 5 custom motor-retention concepts and developing a bayonet-style retainer plus interchangeable 6-in modular avionics bay; manufacturing PETG-CF aerodynamic components with DFM and anisotropic loading to prevent in-flight delamination.
- Calibrating simulations using component-level as-built mass/CG measurements and flight-tested baseline vehicle; converting vehicle loss after successful launch and apogee deployment into revised electronic tracking and high-visibility recovery requirements to prevent future vehicle loss.

## **EXPERIENCE:**

**AIAA | Design, Build, Fly (DBF) Competition:**

*August 2025 – August 2026*

*Structures/Manufacturing Engineer*

- Completed 3 material trade studies comparing 4–6 candidates across bending/torsional strength, mass, cost, and manufacturability to guide wing/fuselage structural material selection.
- Designed tail assembly in SolidWorks intended for 3D printing to reduce part count, enhance manufacturability, and improve aerodynamic performance from previous year.
- Owned wing/fuselage assembly, wood/foam core construction with carbon-fiber reinforcement, and control surface finishing with Monokote heat-shrink covering to reduce drag.

**Bronco S.T.A.R. | BLADE Odyssey:**

*August 2025 – May 2026*

*Communications & Thermals Engineer*

- Developed a heater trade study comparing 3 candidate heating solutions against a ~9.25 W / 37.2 Wh CubeSat power constraint, balancing thermal capability, power draw, size, and integration risk.
- Sized two 60 × 40 mm heating pads at 4.2 V, 3 W each (~0.71 A) within SWaP (size, weight, and power) constraints.
- Produced thermostat-control pseudocode to cycle battery heating between 0–10°C with a 30%-of-flight runtime failsafe for ~4-hour CubeSat mission to maintain safe operating temperature for internal components.

**Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA):**

*September 2025 – November 2025*

*Small Rocket Competition, Team Lead*

- Optimized model rocket fin and body geometries in OpenRocket, SolidWorks, and Bambu Studio to improve projected apogee from 2,100 ft to 2,900 ft (~38% increase).
- Led 5-person design/build team through aerodynamic design, manufacturing, and avionics integration while maintaining project schedule within 2 days of planned milestones.
- Programmed Arduino IDE for avionics systems to read rocket altitude, surrounding pressure, and temperature; post-flight failure analysis identified insufficient epoxy cure time as the cause of internal structural separation.

## **ACADEMIC PROJECTS:**

**ARO1021L** - Analyzed OpenRocket, static-fire, inclinometer, and onboard altimeter apogee estimates across 3 model rockets; quantified method discrepancies and evaluated drag, manufacturing, and environmental sources of error.

**EGR1000L** - Drafted and programmed an Arduino-controlled animatronic system integrating 4 servos and an ~85% 3D-printed structure; iterated actuator selection and mechanical fit to achieve 3 coordinated motions within a 12-in design envelope.

## **SKILLS:**

SolidWorks, Computer-Aided Design, OpenRocket, Ansys Fluent (Basic Proficiency), FDM Additive Manufacturing, Arduino IDE, PrusaSlicer, Carbon Fiber Layups, Wood/Foam Airframe Fabrication, Excel VBA